

Quantum Approach to Solids - Problem Sheet 2 - Phonons

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1. Polythene

Polythene is a polymer made up of $(-\text{CH}_2-)$ units that can be modelled as a 1-D chain of identical masses m , separated by a distance a . The force constant K between the masses is assumed to be identical for all C-C bonds.

Write down the equations of motion for longitudinal displacements of the masses. Show that the characteristic frequencies for this motion along the chain follow the quadratic dispersion relation

$$\omega^2 = \frac{4K}{m} \sin^2\left(\frac{qa}{2}\right),$$

where q is the wavenumber.

2. Polystyrene

Polystyrene is another long chain polymer, with a slightly more complicated repeating unit than polythene. Find out what the repeating unit is for polystyrene. Develop a suitable 1-D model to calculating linear vibrations along polymer chains of polystyrene.

Using this model, calculate the characteristic frequencies of the vibrations of the chain.

Do the relative lengths of the bonds between the two alternating units matter in this calculation. Explain your answer.

By obtaining values for ω as $q \rightarrow 0$ and $q \rightarrow \pm\pi/a$, sketch the dispersion curves for the optical and acoustic branches of the phonon spectrum.

3. Phonon polarization

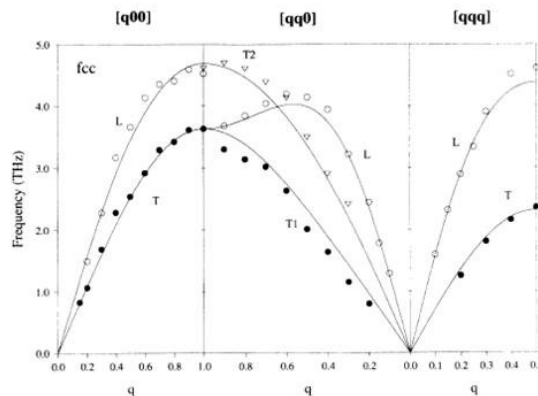
Why can vibrational waves in a solid have more polarization states than vibrational waves in a gas?

4. Vibrations on a periodic lattice

Consider the vibrations of atoms on a 1-D chain, where the atoms are separated by a distance a . Why is the movement of the atoms in the chain the same when multiples of $2\pi/a$ are added to, or subtracted from, the wavenumber?

5. Speed of sound in calcium

The figure below shows calculations and measurements of the phonon spectrum in calcium, an fcc solid. Determine the



sound velocity for longitudinal phonons in the three high symmetry directions shown in the figure. The x-axis is in units of reduced wavevector: $q = 1$ is equivalent to $q = \frac{2\pi}{a}$, where a is the cubic lattice parameter of calcium (this is left to you to find).

6. Umklapp scattering I

What are the differences between normal and Umklapp scattering of phonons? You may find it helpful to sketch both processes in 2D reciprocal space.

7. Phonons in reciprocal space; Umklapp scattering II

A phonon of wavevector $1.3 \times 10^{10} \text{ m}^{-1}$ travelling the [100] direction of a cubic lattice of spacing 0.2 nm combines with another phonon with a wavevector of the same magnitude in the [110] direction. Draw a diagram which shows the magnitude and direction of the resultant phonon in reciprocal space.